#7 Return the heap with the largest k elements. We might as well use heapsort to return the elements in the right order, at a small additional cost.

2.8 More use-cases

The heap is one of the most universally used data structures. Together with stack and queue, they are the basis of almost every algorithm that needs to process the input in a specific order.

Replacing a binary heap with a d-ary heap can improve virtually any piece of code that uses a priority queue. Before delving into a few algorithms that can benefit from the use of a heap, make sure you are familiar with graphs, as most of these algorithms will concern this data structures: to that end, appendix G provides a quick introduction to graphs.

Let's now discuss some algorithms that can benefit from the use of a heap.

2.8.1 Minimum distance in graphs: Dijkstra

Priority queues are crucial to implement Dijkstra and A* algorithms, described in detail in chapter 14, section 14.4 and 14.5; as we will discuss there, the running time of these fundamental algorithms on graphs (which compute the minimum distance to a target) heavily depends on the implementation chosen for the priority queue, and upgrading